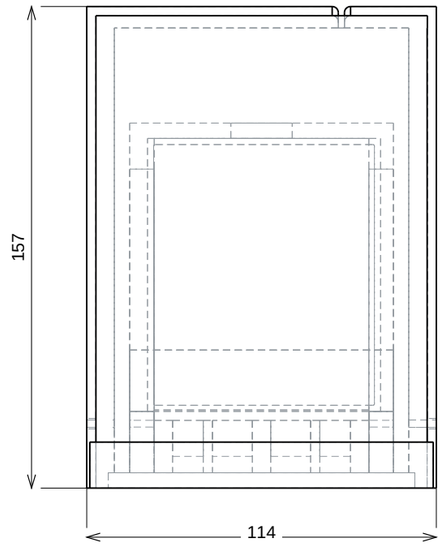
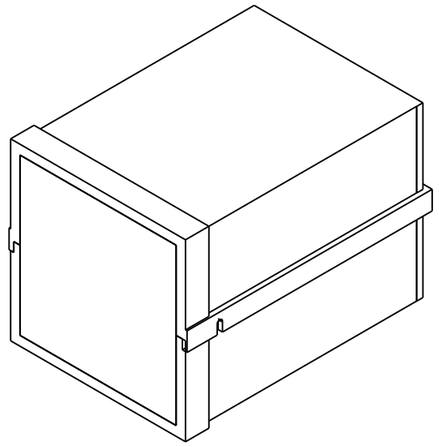
ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

OVERALL ASSEMBLY - FOUR MODELLED COMPONENTS

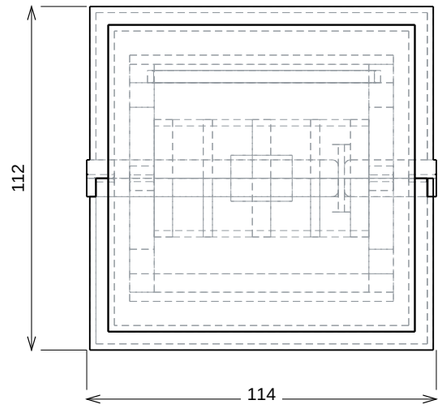
TOP VIEW (+Z)



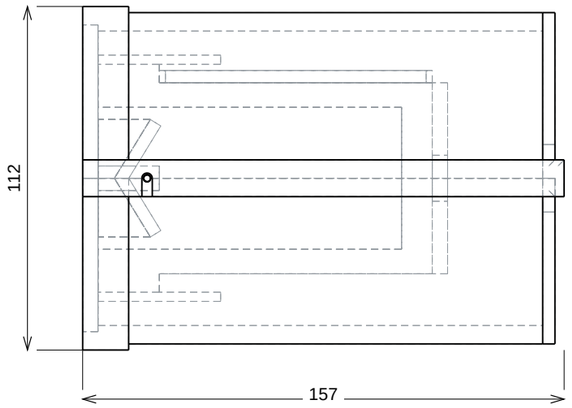
ISOMETRIC - NTS



FRONT VIEW (-Y)



RIGHT SIDE (+X)



ASSEMBLY ENVELOPE

114 X 157 X 112 mm

ITEM	CAD PART NAME	QTY
1	Part2	1
2	Part3	1
3	Part4	1
4	Part4-Mirror3	1

1. Four named instances; all four solids valid.
2. Assembly placements are taken from STEP.
3. No positive-volume part interference found.
4. Camera, LEDs, mirror plates and fasteners are not separate named components here.
5. Hidden edges: thin grey dashed lines.
6. View names follow source assembly axes; they do not establish operating orientation.

Assembly datum A: minimum assembly X/Y/Z.

Source coordinates of A (mm):

X = 623.560207

Y = 834.892308

Z = 1301.120073

Orthographic views: SCALE 1:2.

Isometric view: NTS.

Do not infer clearances from line thickness.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:2

DATE: 16 OCT 2026

DWG: ENG5105-M01

SHEET 1 OF 7

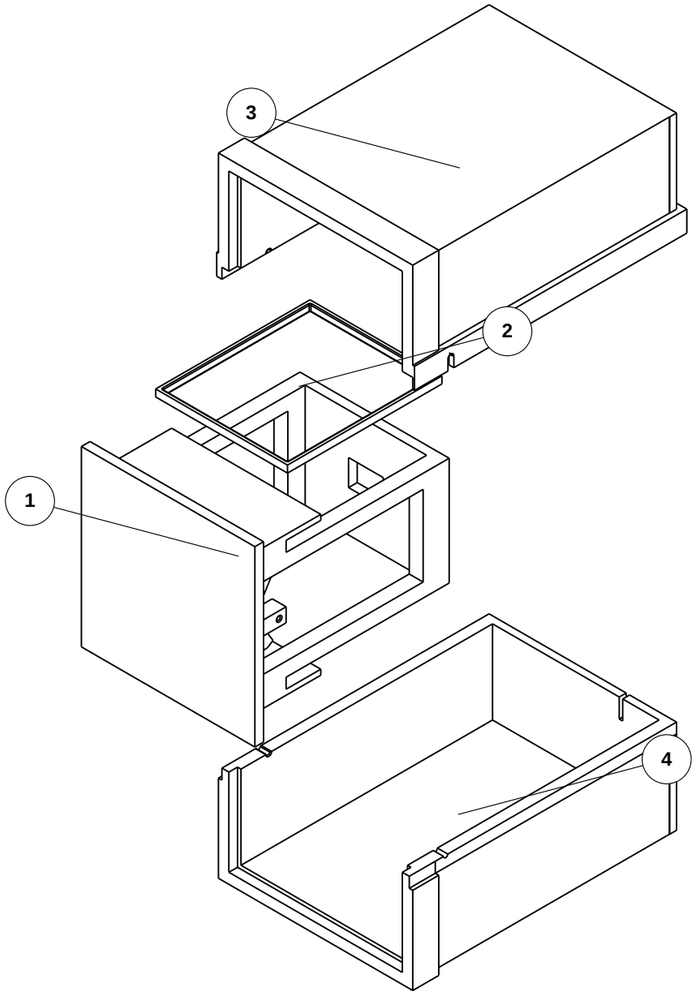
STATUS: REFERENCE

ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

EXPLODED COMPONENT IDENTIFICATION AND POSITION REGISTER

Illustrative offsets only; final assembled locations are preserved in the source STEP.

EXPLODED IDENTIFICATION VIEW



MODEL COMPONENT LIST

ITEM	SOURCE NAME	QTY	ROLE
1	Part2	1	Functional role to confirm
2	Part3	1	Functional role to confirm
3	Part4	1	Functional role to confirm
4	Part4-Mirror3	1	Functional role to confirm

ASSEMBLED BOUNDING-BOX ORIGINS FROM DATUM A

ITEM	Xmin (mm)	Ymin (mm)	Zmin (mm)
1	7.000000	0.000000	6.000000
2	19.817433	25.000000	87.142797
3	0.000000	0.000000	50.000000
4	1.000000	0.000000	0.000000

Offsets locate each part bounding box, not a CAD mate.

Part geometry and orientation are governed by STEP.

Part3 lies in assembly XY; its thickness is along Z.

Exploded shifts (assembly X, Y, Z), mm:

1: (-65, -20, 0) 2: (0, -80, +100)

3: (0, 0, +160) 4: (0, 0, -90)

These shifts are chosen for visibility. They do not

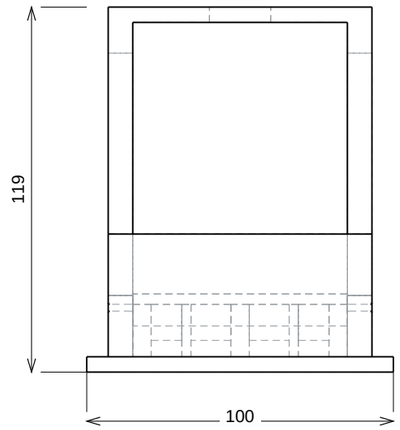
define removal paths, mating features or assembly order.

Fasteners and optical/electrical items are not in this BOM.

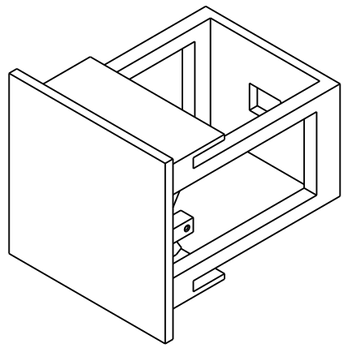
ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

PART2 - DIMENSIONED GEOMETRY REFERENCE

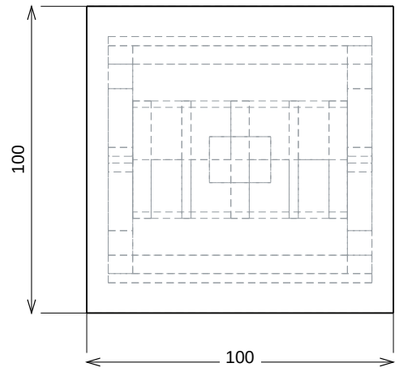
TOP VIEW (+Z)



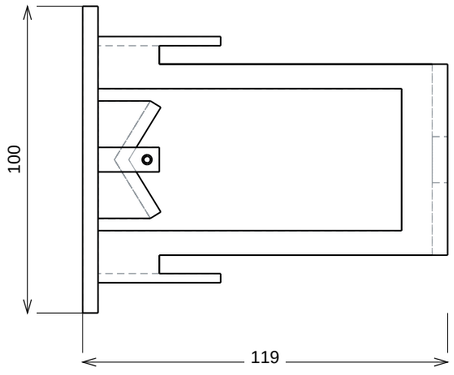
ISOMETRIC - NTS



FRONT VIEW (-Y)



RIGHT SIDE (+X)



GEOMETRY / DIMENSION NOTES

Quantity: 1 | Envelope: 100 X 119 X 100

Full front plate: Y = 0 to 5 (thickness 5).

Rear boundary planes: Y = 114 and 119.

Main inner side planes: X = 15 and 85.

Spacing between these inner planes: 70.

Outer side planes of rails: X = 7 and 93.

V-support faces: 70 span X, 20 slope length.

Included V angle: 117 deg (see sheet 7).

Paired channel axis: Y = 21, Z = 50;

axis parallel to X; cylindrical bore DIA 2.2.

Bore cylinders: STEP faces 2, 5, 30, 57.

Entrances include R0.5 rounded geometry;

entrance arcs reach DIA 3.2.

Do not interpret these bores as threads.

Rear opening boundary planes:

X = 40, 60; Z = 42.5, 57.5.

Opening plane spacing: 20 X 15.

Additional ribs, blends and recesses:

retain the exact supplied STEP geometry.

Material / fit tolerances: not supplied.

LOCAL DATUM: minimum X, Y, Z of this part.

Axes remain parallel to the assembly axes.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:2

DATE: 16 OCT 2026

DWG: ENG5105-M03

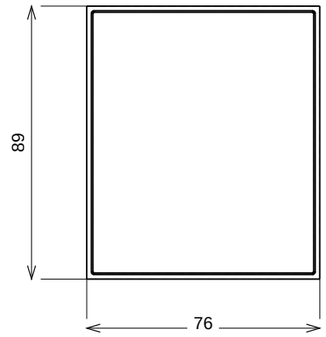
SHEET 3 OF 7

STATUS: REFERENCE

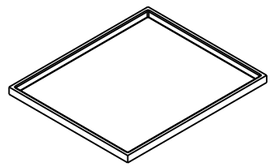
ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

PART3 - DIMENSIONED GEOMETRY REFERENCE

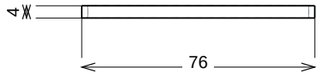
TOP VIEW (+Z)



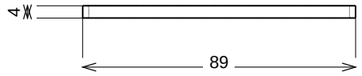
ISOMETRIC - NTS



FRONT VIEW (-Y)



RIGHT SIDE (+X)



GEOMETRY / DIMENSION NOTES

Quantity: 1 | Envelope: 76 X 89 X 4

Part orientation shown as assembled.

Thickness: 4, between Z = 0 and Z = 4.

Inner straight boundary planes:

X = 2 and 74; spacing = 72.

Y = 2 and 87; spacing = 85.

These are face-to-face dimensions,

not a sharp-cornered rectangular opening.

Inner boundary includes rounded features.

Small cylindrical edge blends: R0.5.

Other rounded corner arcs: R1.0.

Nominal straight frame width: 2.

No separate threaded hardware is modelled.

No camera PCB / lens is present as a part.

Function and attachment method: to confirm.

Use STEP for the full corner transitions.

Material / fit tolerances: not supplied.

LOCAL DATUM: minimum X, Y, Z of this part.

Axes remain parallel to the assembly axes.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:2

DATE: 16 OCT 2026

DWG: ENG5105-M04

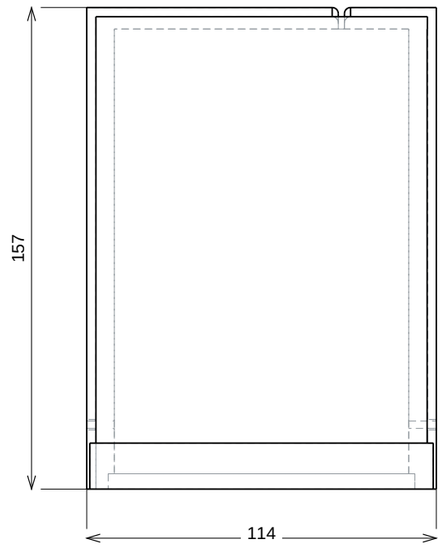
SHEET 4 OF 7

STATUS: REFERENCE

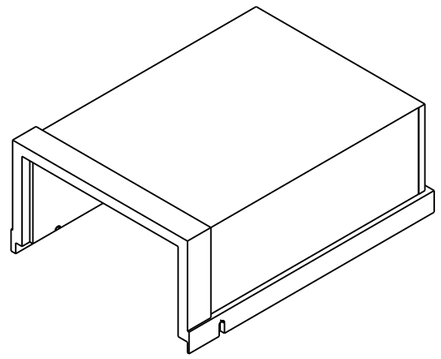
ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

PART4 - DIMENSIONED GEOMETRY REFERENCE

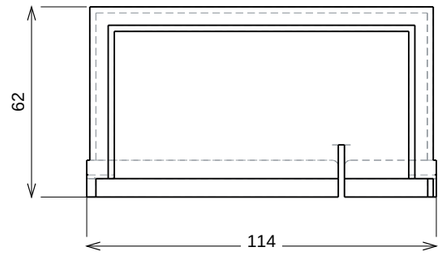
TOP VIEW (+Z)



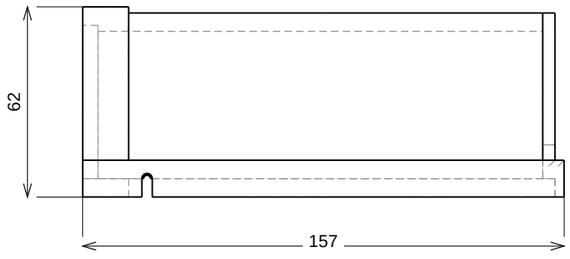
ISOMETRIC - NTS



FRONT VIEW (-Y)



RIGHT SIDE (+X)



GEOMETRY / DIMENSION NOTES

Quantity: 1 | Envelope: 114 X 157 X 62

Main left wall: X = 3 to 9; thickness 6.

Main right wall: X = 105 to 111; thickness 6.

Inner plane spacing along X: 96.

Main roof: Z = 54 to 60; thickness 6.

Rear wall planes: Y = 150 and 154.

Local rear extension reaches Y = 157.

Front aperture boundary planes:

X = 7, 107; Z = 6, 56.

Boundary-plane spacing: 100 X 50.

Side channel axis: Y = 21, Z = 6;

axis parallel X; cylinder radius 1.2.

Channel is locally trimmed by mating edges;

DIA 2.4 describes the generating cylinder.

Selected rear cutout arcs: R2.0.

Arc axis X coordinates: 80.035343,

86.035343; see geometry register.

Do not assign thread sizes to these cuts.

Material / fit tolerances: not supplied.

LOCAL DATUM: minimum X, Y, Z of this part.

Axes remain parallel to the assembly axes.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:2

DATE: 16 OCT 2026

DWG: ENG5105-M05

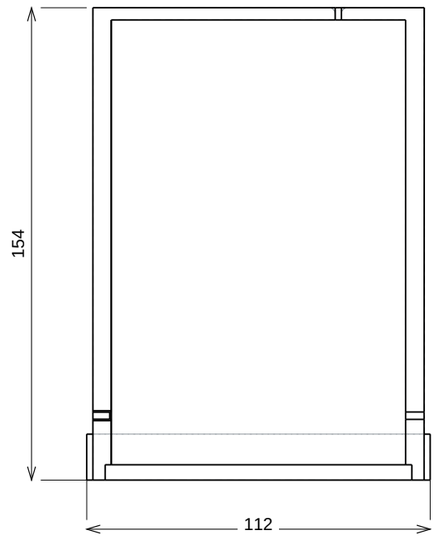
SHEET 5 OF 7

STATUS: REFERENCE

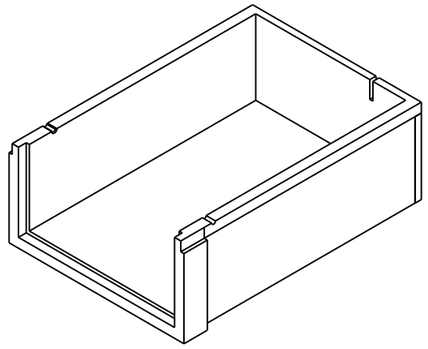
ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

PART4-MIRROR3 - DIMENSIONED GEOMETRY REFERENCE

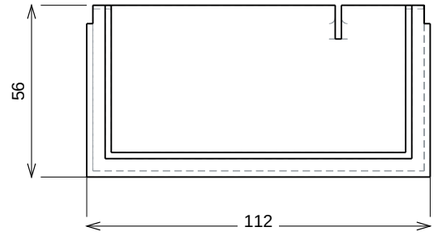
TOP VIEW (+Z)



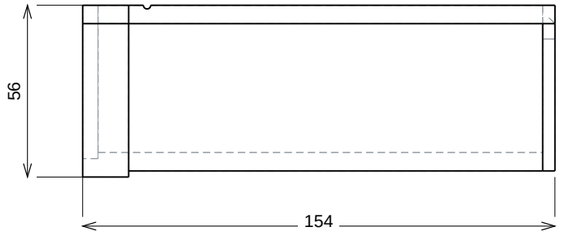
ISOMETRIC - NTS



FRONT VIEW (-Y)



RIGHT SIDE (+X)



GEOMETRY / DIMENSION NOTES

Quantity: 1 | Envelope: 112 X 154 X 56

Name is retained verbatim from source CAD.

No optical mirror plate is inferred from name.

Main left wall: X = 2 to 8; thickness 6.

Main right wall: X = 104 to 110; thickness 6.

Inner plane spacing along X: 96.

Main base: Z = 2 to 8; thickness 6.

Rear wall planes: Y = 150 and 154.

Front aperture boundary planes:

X = 6, 106; Z = 6, 56.

Boundary-plane spacing: 100 X 50.

Side channel axis: Y = 21, Z = 56;

axis parallel X; cylinder radius 1.2.

Channel is an edge-trimmed cylindrical cut,

not a specified threaded mounting hole.

Selected rear cutout arcs: R2.0.

Arc axis X coordinates: 79.035343,

85.035343; see geometry register.

Material / fit tolerances: not supplied.

LOCAL DATUM: minimum X, Y, Z of this part.

Axes remain parallel to the assembly axes.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:2

DATE: 16 OCT 2026

DWG: ENG5105-M06

SHEET 6 OF 7

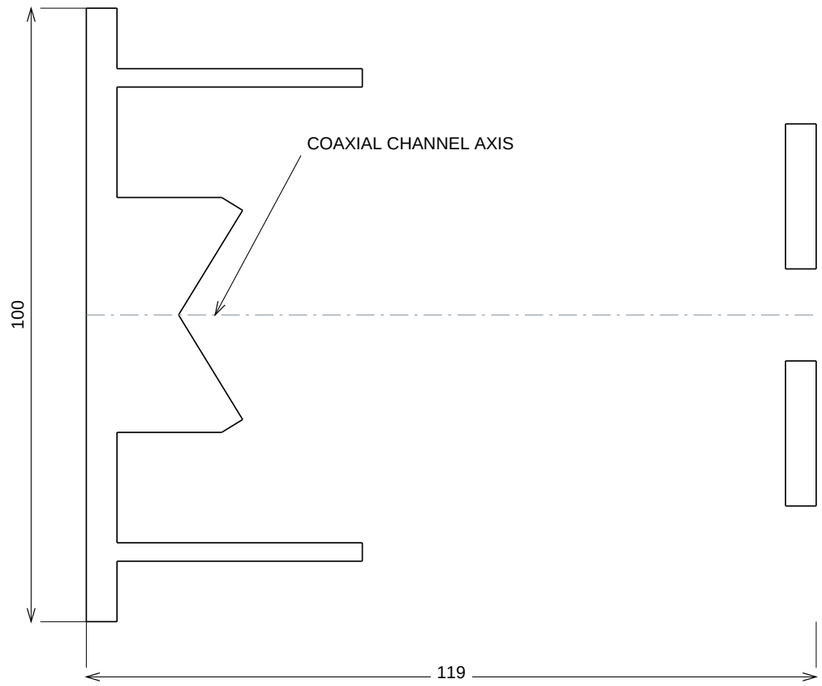
STATUS: REFERENCE

ONLINE QUALITY-MONITORING SYSTEM FOR RECYCLED PLA FILAMENT

OPTICAL MOUNTING GEOMETRY - VERIFIED SUPPORT FACES

Part2 reference section only. Separate mirror plates, camera optics and filament are not modelled.

SECTION AT PART2 LOCAL X = 50

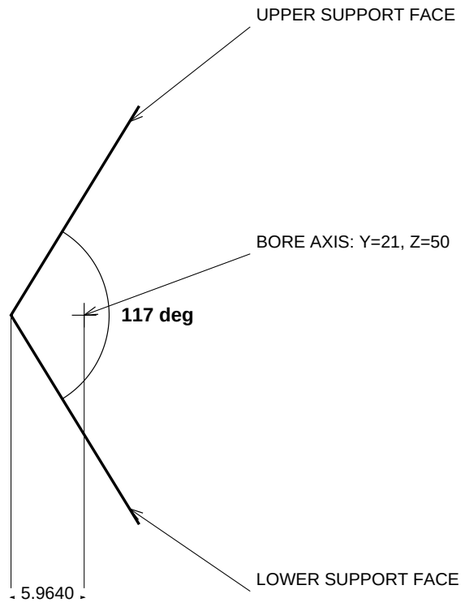


Section edges are computed from the exact STEP solid.

Local Y is horizontal; local Z is vertical.

DIA 2.2 bores lie in side rails; X = 50 passes through the open V.

V-SUPPORT DETAIL - SCALE 2:1



Each support face: 70 along X X 20 along slope.

Face area: 1400 mm2 each. V apex: Y=15.035996, Z=50.

117 deg is the included angle between support rays (+Y).

Face-normal angle is 63 deg; it is not the included V angle.

ASSEMBLED AXIS LOCATION

From assembly datum A:

Y = 21.000000 mm

Z = 56.000000 mm

Direction: parallel to assembly X.

The Part2 DIA 2.2 bore axis
aligns with the R1.2 cylindrical
cuts in Part4 and Part4-Mirror3.

NOT RECOVERED FROM THIS CAD:

- Actual mirror plate dimensions
- Mirror thickness / adhesive layer
- Camera optical centre / axis
- Lens-to-filament working distance
- Actual filament path confirmation
- LED and diffuser placement

The 70 X 20 dimensions describe
support geometry, not mirror stock.
No ray paths or optical distances
are assumed on this drawing.

MONASH UNIVERSITY | ENG5105 FINAL ELECTRICAL DESIGN REPORT

SOURCE: Assem1.STEP | All dimensions in mm | CAD REFERENCE - REV A

Unshown geometry: supplied STEP governs. Materials and tolerances not specified.

Views use assembly axes. Drawing scale applies when printed at 100% on A3.

PROJECTION: THIRD ANGLE

SCALE: 1:1 / DETAIL 2:1

DATE: 16 OCT 2026

DWG: ENG5105-M07

SHEET 7 OF 7

STATUS: REFERENCE